

Econ 140 Presentation

Do NBA Forwards get a Higher Salary than other Positions in the NBA?



Our Hypothesis

- We believed that NBA Forwards get a higher salary than Non-Forward positions
- NBA positions include Point Guard, Shooting Guard, Small Forward, Power Forward, and Center
- We believed that Forwards had higher salary because NBA Forwards tend to be the most versatile players on a team
- Forwards typically have the most diverse skill sets as they can play on all areas of the court due to the desired size to play forward, desired talents to play forward, and their mix between Guard and Center skill traits
- Examples of NBA forwards include LeBron James, Scottie Pippen, Jayson Tatum, Kevin Durant, and Paul Pierce



Dataset

- We analyzed a library from the Woolridge package called 'nbasal'
- Our outcome variable was 'lwage,' which is the natural log of the players' incomes and our treatment variable of interest was 'forward' which is a binary variable representing whether the player is a forward or not
- For our first regression, we looked at these two variables along with a few controls

▶ head(nbasal)



A data.frame: 6 × 22

	marr	wage	exper	age	coll	games	minutes	guard	forward	center	...	assists	draft	allstar	avgmin	lwage	black	children	expersq	agesq	marrblack
	<int>	<dbl>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	...	<dbl>	<int>	<int>	<dbl>	<dbl>	<int>	<int>	<int>	<int>	<int>
1	1	1002.5	4	27	4	77	2867	1	0	0	...	4.5	19	0	37.23376	6.910252	1	0	16	729	1
2	1	2030.0	5	28	4	78	2789	1	0	0	...	8.8	28	0	35.75641	7.615791	1	1	25	784	1
3	0	650.0	1	25	4	74	1149	0	0	1	...	0.2	19	0	15.52703	6.476973	1	0	1	625	0
4	0	2030.0	5	28	4	47	1178	0	1	0	...	1.5	1	0	25.06383	7.615791	1	0	25	784	0
5	0	755.0	3	24	4	82	2096	1	0	0	...	2.6	24	0	25.56098	6.626718	1	0	9	576	0
6	0	2014.5	9	31	4	82	1971	0	1	0	...	1.5	4	0	24.03658	7.608126	1	0	81	961	0

Short Regression Results

```
▶ short_reg <- lm(lwage ~ forward + age + coll + black + agesq, data = nbasal)
summary(short_reg)
```

```
Call:
lm(formula = lwage ~ forward + age + coll + black + agesq, data = nbasal)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.4637	-0.4074	0.1181	0.5574	1.8574

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.707435	2.682441	0.264	0.7922
forward	0.131853	0.103129	1.279	0.2022
age	0.366551	0.188392	1.946	0.0528 .
coll	-0.101227	0.067254	-1.505	0.1335
black	0.308441	0.128549	2.399	0.0171 *
agesq	-0.004886	0.003272	-1.494	0.1365

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8246 on 263 degrees of freedom
Multiple R-squared: 0.1409, Adjusted R-squared: 0.1246
F-statistic: 8.629 on 5 and 263 DF, p-value: 1.376e-07

- In our short regression, being a forward was associated with approximately a 13.2% increase in wage; however, there was no statistical significance
- So, we decided to look for some omitted variables
- We decided on three
- 'exper'
 - How many years player has played professionally
- 'points'
 - The points per game by a player
- 'games'
 - The average amount of games a player has played per season

‘exper’ long reg/aux reg

```
exper_long_reg <- lm(lwage ~ forward + exper + age + coll + black + agesq, data = nbasal)
summary(exper_long_reg)
```



```
Call:
lm(formula = lwage ~ forward + exper + age + coll + black + agesq,
    data = nbasal)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.41855 -0.38266  0.09234  0.49506  1.93049
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.943141   2.577920   1.530   0.1273
forward      0.143808   0.096929   1.484   0.1391
exper        0.251931   0.042077   5.987 7e-09 ***
age          0.280518   0.177610   1.579   0.1154
coll        -0.122601   0.063298  -1.937   0.0538 .
black        0.214365   0.121812   1.760   0.0796 .
agesq       -0.007535   0.003106  -2.426   0.0159 *
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.7749 on 262 degrees of freedom
Multiple R-squared:  0.2443, Adjusted R-squared:  0.227
F-statistic: 14.12 on 6 and 262 DF, p-value: 6.331e-14
```

- In our long regression, we see that ‘forward’ is associated with approximately a 14.4% increase in wage; however, again, this is not statistically significant
- We do see that ‘exper’ is statistically significant though and is associated with approximately a 25.2% increase in wage
- When we go to our auxiliary regression though we see that there is no statistical significance, meaning that we are not able to say that leaving out experience causes omitted variable bias
- Next, we looked at ‘points’

```
exper_aux_reg <- lm(exper ~ forward + age + coll + black + agesq, data = nbasal)
summary(exper_aux_reg)
```



```
Call:
lm(formula = exper ~ forward + age + coll + black + agesq, data = nbasal)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-3.9137 -0.6739  0.1446  0.7621  3.2394
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.843609   3.693909  -3.477 0.000593 ***
forward      -0.047452   0.142016  -0.334 0.738548
age          0.341492   0.259428   1.316 0.189211
coll         0.084841   0.092613   0.916 0.360468
black        0.373420   0.177020   2.109 0.035849 *
agesq        0.010514   0.004505   2.334 0.020361 *
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.136 on 263 degrees of freedom
Multiple R-squared:  0.8905, Adjusted R-squared:  0.8885
F-statistic: 427.9 on 5 and 263 DF, p-value: < 2.2e-16
```


'points' long reg/aux reg

```
points_long_reg <- lm(lwage ~ forward + points + age + coll + black + agesq, data = nbasal)
summary(points_long_reg)
```

Call:
lm(formula = lwage ~ forward + points + age + coll + black + agesq, data = nbasal)

Residuals:

	Min	1Q	Median	3Q	Max
	-1.8717	-0.3650	0.1135	0.4616	1.7064

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.908134	2.131622	1.364	0.1736
forward	0.160847	0.081707	1.969	0.0501 .
points	0.087139	0.006947	12.543	<2e-16 ***
age	0.148050	0.150212	0.986	0.3252
coll	-0.004528	0.053818	-0.084	0.9330
black	0.140446	0.102683	1.368	0.1726
agesq	-0.001396	0.002606	-0.536	0.5926

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6531 on 262 degrees of freedom
Multiple R-squared: 0.4632, Adjusted R-squared: 0.4509
F-statistic: 37.69 on 6 and 262 DF, p-value: < 2.2e-16

- In our long regression, we see that 'forward' is associated with approximately a 16.1% increase in wage; however, again, this is not statistically significant
- 'points' is statistically significant here, being associated with approximately a 9% increase in wage
- Like our last auxiliary regression, there is again no statistical significance so we are not able to draw conclusions
- Our next possible omitted variable was 'games'

```
points_aux_reg <- lm(points ~ forward + age + coll + black + agesq, data = nbasal)
summary(points_aux_reg)
```

Call:
lm(formula = points ~ forward + age + coll + black + agesq, data = nbasal)

Residuals:

	Min	1Q	Median	3Q	Max
	-11.4567	-4.5581	-0.5317	3.8363	20.1788

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-25.25516	18.85563	-1.339	0.1816
forward	-0.33273	0.72492	-0.459	0.6466
age	2.50751	1.32426	1.894	0.0594 .
coll	-1.10971	0.47275	-2.347	0.0196 *
black	1.92791	0.90360	2.134	0.0338 *
agesq	-0.04006	0.02300	-1.742	0.0827 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.797 on 263 degrees of freedom
Multiple R-squared: 0.05296, Adjusted R-squared: 0.03495
F-statistic: 2.941 on 5 and 263 DF, p-value: 0.01331

'games' long reg/aux reg

```
games_long_reg <- lm(lwage ~ forward + games + age + coll + black + agesq, data = nbasal)
summary(games_long_reg)
```

Call:

```
lm(formula = lwage ~ forward + games + age + coll + black + agesq,
    data = nbasal)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.1977	-0.4284	0.1080	0.5674	2.0782

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.630093	2.539207	0.642	0.5215
forward	0.132549	0.097425	1.361	0.1748
games	0.014737	0.002577	5.718	2.93e-08 ***
age	0.244283	0.179252	1.363	0.1741
coll	-0.088175	0.063575	-1.387	0.1666
black	0.225095	0.122310	1.840	0.0668 .
agesq	-0.002948	0.003109	-0.948	0.3439

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.779 on 262 degrees of freedom
Multiple R-squared: 0.2362, Adjusted R-squared: 0.2188
F-statistic: 13.51 on 6 and 262 DF, p-value: 2.389e-13

```
games_aux_reg <- lm(games ~ forward + age + coll + black + agesq, data = nbasal)
summary(games_aux_reg)
```

Call:

```
lm(formula = games ~ forward + age + coll + black + agesq, data = nbasal)
```

Residuals:

Min	1Q	Median	3Q	Max
-64.812	-6.159	7.002	12.407	21.734

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-62.60714	60.62969	-1.033	0.3027
forward	-0.04722	2.33096	-0.020	0.9839
age	8.29651	4.25811	1.948	0.0524 .
coll	-0.88563	1.52010	-0.583	0.5607
black	5.65550	2.90551	1.946	0.0527 .
agesq	-0.13151	0.07394	-1.778	0.0765 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 18.64 on 263 degrees of freedom
Multiple R-squared: 0.04063, Adjusted R-squared: 0.02239
F-statistic: 2.228 on 5 and 263 DF, p-value: 0.05198

- Here, in our long regression, we see that forward is now associated with approximately a 13.3% increase in wage; however, with no statistical significance
- 'games' is associated with approximately a 1.5% increase in wage and is statistically significant
- Again, we also see no statistical significance in our auxiliary regression so no conclusions are made

Instrumental Variable

- After looking at multiple variables to check for OVB, we decided to take another approach, IV regression
- We decided to use the variable 'draft,' which is the player's draft position, as our instrumental variable because it has some randomness to it
- The reason for this is that a teams position, if they are not in the playoffs, are given a pick in the draft based on a lottery
- Additionally, players are not always chosen based on whether they are the best, but instead what position the team is looking to fill
- However, we understood that the exogeneity and exclusion restriction were not satisfied totally here, we wanted to take a chance to look at the results
 - The reason for this is that we know that draft position is impacted by how good of player you are before the draft
 - We also know that draft number can impact wage directly, because higher drafts are usually more talented leading to being a better player and a higher wage

IV Regression Results

```
iv_reg <- ivreg(lwage ~ forward
               + age + coll + black + agesq
               | draft + age + coll + black + agesq,
               data = nbasal)

summary(iv_reg)
```

Call:
ivreg(formula = lwage ~ forward + age + coll + black + agesq |
draft + age + coll + black + agesq, data = nbasal)

Residuals:

Min	1Q	Median	3Q	Max
-5.110	-2.407	1.131	2.090	3.464

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.106968	8.357965	0.850	0.3960
forward	4.668004	2.070102	2.255	0.0251 *
age	-0.189301	0.614024	-0.308	0.7581
coll	0.001204	0.232173	0.005	0.9959
black	-0.187322	0.455290	-0.411	0.6811
agesq	0.004467	0.010656	0.419	0.6755

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	1	234	5.495	0.0199 *
Wu-Hausman	1	233	61.276	1.73e-13 ***
Sargan	0	NA	NA	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.366 on 234 degrees of freedom
Multiple R-Squared: -7.36, Adjusted R-squared: -7.539
Wald test: 1.842 on 5 and 234 DF, p-value: 0.1054

- In our IV being a forward is associated with a 466% increase in wage
 - This is way larger than we would ever expect proving how draft number is most likely not a great instrumental variable
- These results confirm our ideas before about how draft position does not satisfy the IV assumptions

Final Conclusions

- Not an RCT, so we can not accept or deny causality as there is selection bias
- Hard to make a causal study as we can't randomize positions, which players play on which teams, draft order, how many points are scored, etc
- Our variables were statistically significant but there was not statistical significance in the auxiliary, which means there is no omitted variable bias. Thus players get paid more not due to their position, but more because of their points scored, games played, and experience. And being a forward does not increase these variables by a statistically significant margin
- We fail to reject the null hypothesis



References

"Wooldridge: Data Sets for Introductory Econometrics." *CRAN*, Comprehensive R Archive Network, <https://cran.r-project.org/web/packages/wooldridge/wooldridge.pdf>. Accessed 4 Dec. 2024.

THANK YOU